

# Marcell Howard

## PERSONAL DETAILS

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## RESEARCH INTERESTS

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Gravitational Particle Production, Early Universe Cosmology, Cosmological Gravitational Wave Backgrounds, Quantum Cosmology, Primordial Black Holes

## ACADEMIC APPOINTMENTS

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**Postdoctoral Research Associate** *2026 - Present*  
*Brown University*  
Advised by Stephon Alexander

## EDUCATION

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**University of Pittsburgh, Pittsburgh, PA** *2019 - 2026*  
*Ph.D., Physics*

**Case Western Reserve University, Cleveland, OH** *2015 - 2019*  
*B.S., Mathematics and Physics*

## RESEARCH EXPERIENCE

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**Graduate Research Assistant** *2020 - 2026*  
*University of Pittsburgh advised by Arthur Kosowsky*  
Doctoral Thesis: Issues of Gravitation in Cosmology

**Senior Project** *2018 - 2019*  
*Case Western Reserve University advised by Kurt Hinterbichler*  
Senior Project Thesis: The People vs Mimetic Gravity

**Research Experience for Undergraduates** *Summer 2018*  
*Rutgers, the State University of New Jersey advised by Andrew Baker*  
Probing the Evolution of Galaxies Using Stellar Mass Selected Samples

**Research Experience for Undergraduates** *Summer 2017*  
*Cornell University advised by Gordon Stacey*  
Using Far-Infrared Fine-Structure Lines for Characterizing Star Formation Processes in Nearby Galaxies

## FELLOWSHIPS, SCHOLARSHIPS AND AWARDS

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### PITT PACC Fellowship

*Summer 2026*

Competitive internal department fellowship for excellence in research and scholarship

### dB-SERC Mentor-Mentee Evidence-Based Teaching Award

*Fall 2025*

Competitive teaching award for evidence-based teaching with a focus on transforming at least a two-week segment of a natural science course to improve student learning outcomes in a measurable way

### Provost's Dissertation Completion Fellowship - Fall 2025

*Declined*

Competitive university-wide fellowship that acknowledges academic and scholarly accomplishments and ensure dedicated time to focus full-time on finishing the dissertation

### Peter F.M. Koehler Pre-Doctoral Fellowship

*2023 - 2024*

Competitive internal department fellowship for excellence in teaching and research

### 2021 Honorable Mention, Ford Foundation Predoctoral Fellowship

*2021*

### Dietrich School of Arts and Sciences Summer Research Predoctoral Fellowship

*2020*

Fellowship to support the most well-prepared students and speed the transition into the research that will become thesis work and to help meet diversity goals

### Kenneth P. Dietrich School of Arts & Sciences Fellowship

*2019*

For incoming graduate students for recognition of an outstanding undergraduate record

### Case Alumni Junior/Senior Scholarship

*2017 - 2019*

For rising sophomores and juniors pursuing a B.S. in engineering, math, or the applied sciences based on merit, need and personal skills

## PUBLICATIONS LIST

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\* = *Corresponding Author*

- M. Howard\* and M. König, *Elastic scattering of cosmological gravitational wave backgrounds: primordial black holes and stellar objects*, *J. Cosmology Astropart. Phys.* **2024** (2024) 045 [[2309.15925](#)]
- T. Daniel, M. Howard\* and M. König, *An SZ-like effect on cosmological gravitational wave backgrounds*, *J. Cosmology Astropart. Phys.* **2023** (2023) 041 [[2308.00111](#)]
- S. Alexander, T. Daniel, M. Howard\* and M. König, *Exact fermionic Chern-Simons-Kodama state in quantum gravity*, *Phys. Rev. D* **106** (2022) 106012 [[2207.11856](#)]
- M. Howard\*, A. Kosowsky and G. Valogiannis, *Galaxy Cluster Statistics in Modified Gravity Cosmologies*, *arXiv e-prints* (2022) [arXiv:2205.13015](#) [[2205.13015](#)]

## SEMINARS AND COLLOQUIA

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|----------------------------------------------------------------|------------------|
| University of Maryland, EPT Seminar                            | <i>Nov 2025</i>  |
| Indiana University Bloomington, HEP Seminar                    | <i>Nov 2025</i>  |
| Rice University, Astronomy Undergraduate Seminar               | <i>Oct 2025</i>  |
| University of Florida, HEP Seminar                             | <i>Oct 2025</i>  |
| University of Pittsburgh, Mathematics Graduate Student Seminar | <i>Sep 2025</i>  |
| University of Illinois Urbana-Champaign, HEP Seminar           | <i>Sep 2025</i>  |
| University of Pittsburgh, AstroLunch                           | <i>Jan 2025</i>  |
| Pomona College Colloquium                                      | <i>Sep 2024</i>  |
| Simons-NSBP Scholars Program, Flatiron Institute               | <i>July 2024</i> |
| Syracuse University Colloquium                                 | <i>Nov 2023</i>  |
| Syracuse University, Watson Cosmology Group Seminar            | <i>Nov 2023</i>  |
| MIT, String Theory Group Meeting                               | <i>Feb 2023</i>  |
| Case Western Reserve University, Particle Astrophysics Seminar | <i>Sep 2022</i>  |
| Brown University, Alexander Lab Group Meeting                  | <i>Mar 2022</i>  |

## SELECTED WORKSHOPS AND CONFERENCES

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<sup>†</sup> = *Invited Talk*

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|-------------------------------------------------------------------------------------------------------|------------------|
| 33rd Texas Symposium on Relativistic Astrophysics                                                     | <i>Dec 2025</i>  |
| 28th International Conference on Particle Physics and Cosmology                                       | <i>Oct 2025</i>  |
| Division of Particles & Fields - Phenomenology Symposium 2024                                         | <i>May 2024</i>  |
| <sup>†</sup> International Symposium on Particles, Strings, and Cosmology                             | <i>June 2023</i> |
| <sup>†</sup> Ending Inflation and the Hot Big Bang Workshop<br>Simons Center for Geometry and Physics | <i>June 2023</i> |
| Pennsylvania State University, Neighborhood Workshop                                                  | <i>Apr 2023</i>  |

American Physical Society April Meeting 2022

Apr 2022

National Society of Black Physicists Conference

Nov 2021

## TEACHING EXPERIENCE

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### Course Instructor

*PHYS 0110: Introduction to Physics 1*

### Guest Lecturer

*PHYS 3726: General Relativity 2*

General Relativity as a Quantum Field Theory

### Graduate Teaching Assistant

*PHYS 0111: Introduction to Physics 2*

*PHYS 0174: Basic Physics for Science and Engineering 1*

*PHYS 0175: Basic Physics for Science and Engineering 2*

*PHYS 0212: Introduction to Laboratory Physics*

*PHYS 0476: Basic Physics for Science and Engineering II - Honors*

- Held two-four recitation sessions per week.
- Graded quizzes and tests.

### Graduate Core Course Departmental Tutor

*PHYS 2565: Non-Relativistic Quantum Mechanics 1*

*PHYS 2566: Non-Relativistic Quantum Mechanics 2*

*PHYS 1373 & 2373 (Combined section): Mathematical Methods in Physics*

- Held two office hours per week.
- Provided homework help as well as furthered conceptual understanding of the material.

### Undergraduate Supplemental Instructor

*MATH 121: Calculus for Scientists and Engineers I*

*MATH 122: Calculus for Scientists and Engineers II*

- Held three small groups and one recitation session per week.
- Graded quizzes and tests.

### Undergraduate Teaching Assistant

*PHYS 121: General Physics I - Mechanics*

*PHYS 122: General Physics II - Electricity and Magnetism*

- Graded written homework handed in every week.
- Handed out in-class assignments and provided help to various students.

## OUTREACH AND SERVICE

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|                                                                                                                                              |                    |
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| <b>Allegheny Observatory Public Lecture Series</b><br><i>Allegheny Observatory</i><br>Invited Presenter: The First Three Minutes             | <i>July 2025</i>   |
| <b>Exploring the Universe with Pitt Astronomers</b><br><i>University of Pittsburgh</i><br>Volunteer Presenter                                | <i>Aug 2024</i>    |
| <b>Astronomy on Tap</b><br><i>Two Fray's Brewery</i><br>Volunteer Presenter: When Black Holes Aren't So Black                                | <i>Aug 2024</i>    |
| <b>Astrophysicist for a Day</b><br><i>Allegheny Observatory</i><br>Volunteer Workshop Leader: Rediscovering Our Accelerating Universe        | <i>Mar 2024</i>    |
| <b>Society of Physics Students Graduate Student Panel</b><br><i>University of Pittsburgh</i><br>Panelist                                     | <i>Dec 2023</i>    |
| <b>Python Boot Camp 2023: Functions and Modules</b><br><i>University of Pittsburgh</i><br>Volunteer Presenter                                | <i>May 2023</i>    |
| <b>Astronomy on Tap NYC</b><br><i>Pete's Candy Store</i><br>Invited Presenter: The Wave function of the Universe                             | <i>May 2023</i>    |
| <b>International Baccalaureate Class Group Project</b><br><i>St Edward High School</i><br>Invited Speaker                                    | <i>Mar 2023</i>    |
| <b>Seminar at High School (SAHS) and Pathway Career Speaker Series</b><br><i>Woodland Hills High School</i><br>Invited Speaker               | <i>Dec 2022</i>    |
| <b>Society of Physics Students Graduate Student Panel</b><br><i>University of Pittsburgh</i><br>Panelist                                     | <i>Oct 2022</i>    |
| <b>Astronomy on Tap</b><br><i>Two Fray's Brewery</i><br>Volunteer Presenter: The Wave function of the Universe                               | <i>Sept 2022</i>   |
| <b>Summer Seminar Series: So You Wanna Get Good At L<sup>A</sup>T<sub>E</sub>X</b><br><i>University of Pittsburgh</i><br>Volunteer Presenter | <i>July 2022</i>   |
| <b>Cornell's Focus for Teens</b><br><i>Cornell University</i><br>Workshop Leader                                                             | <i>Summer 2017</i> |

## MEDIA CONTRIBUTIONS AND APPEARANCES

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| <b>Black In Cosmology</b><br><i>Nature Astronomy</i><br><a href="https://doi.org/10.1038/s41550-025-02577-9">https://doi.org/10.1038/s41550-025-02577-9</a> | <i>June 2025</i> |
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## WINTER AND SUMMER SCHOOLS

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39th Advanced School in Theoretical Physics on Geometry, Topology and  
Mechanics in Soft Condensed Matter

*Jan 2024*

*Israel Institute for Advanced Study*

Canceled due to war

## SKILLS

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*Programming*

Python, Mathematica, Matlab

*Languages*

*Software*

SciPy/matplotlib, AstroPy, git

## PROFESSIONAL ORGANIZATIONS

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Pittsburgh Particle Astrophysics and Cosmology Center

*2020 - 2026*

National Society of Black Physicists

*2018 - Present*

## REFERENCES

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Arthur Kosowsky

[kosowsky\[at\]pitt.edu](mailto:kosowsky[at]pitt.edu)

University of Pittsburgh

Andrew Zentner

[zentner\[at\]pitt.edu](mailto:zentner[at]pitt.edu)

University of Pittsburgh

Stephon Alexander

[ariel\\_green\[at\]brown.edu](mailto:ariel_green[at]brown.edu)

Brown University